

Materials Thermodynamics and Kinetics



MT-26002: Materials Thermodynamics and Kinetics (PCC)

2nd Law of Thermodynamics

- In any spontaneous or natural process, the total entropy of an isolated system always increases over time
- Look at the 1st Law
- It tells us energy cannot be created or destroyed—only converted
- Consider a rock – it can spontaneously launch itself into the air by absorbing thermal energy from the ground – no violation of conservation of energy



Entropy

- But metallurgical reactions or extractions happen on their own, and they need spontaneity
- Here comes the term ***ENTROPY***
- And this provides the 2nd law of thermodynamics correlated with nature



- A chemical reaction is feasible at constant temperature and pressure if the change in enthalpy is negative
- $Sn (tin)_{white, 298 K} \rightarrow Sn (tin)_{grey, 298 K}$
- $\Delta H = -500 \text{ cal}$
- Feasible ???



- But stable form is $Sn (tin)_{white, 298 K}$
- There should be a thermodynamic property that can support this spontaneity
- That is known as **ENTROPY**
- It describes the order of the system
- Boltzmann proposed $S = k \ln W + S_0$

$k = R/N_A$, W is thermodynamic probability of the system, S_0 is a constant



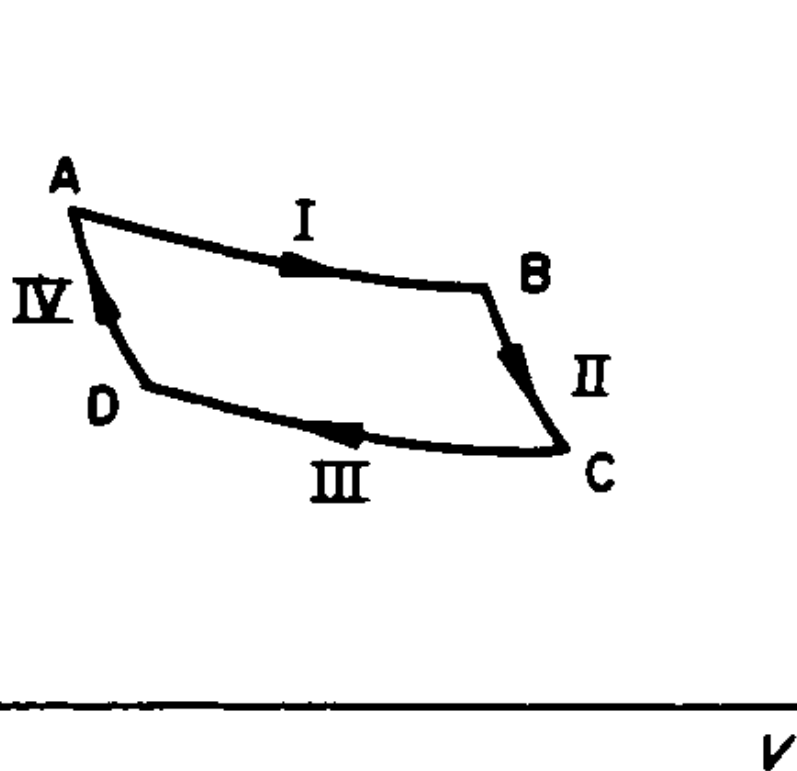
- $S = k \ln W$

When S_0 is 0 at $W = 1$

- Entropy (S) is a state function
- Spontaneity is related to the entropy of the system and its surroundings
- Solid $\rightarrow$ Liquid $\rightarrow$ Gas



Carnot Cycle



- Clausius: Heat absorbed by a system (q) to the temperature at which it is absorbed (T), the entropy change of the system is $dS = \frac{\delta q}{T}$



Derivations

- Variation of entropy with temperature (constant volume)
- Constant q (adiabatic)
- If the pressure and volume are not constant, for a small heat change
- Variation of entropy with volume (isothermal)
- For a reversible process

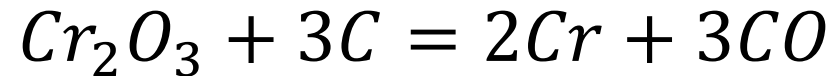


Combined Statements of 1st & 2nd Laws



Problem

Calculate the standard entropy change for the reaction



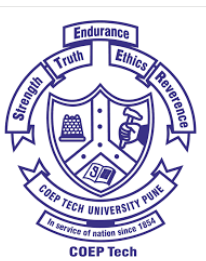
at 25 °C (298 K).

Given: $S_{298, \text{Cr}_2\text{O}_3}^0 = 19.4 \frac{\text{cal}}{\text{deg.mole}}$

$$S_{298, \text{C}}^0 = 1.36 \frac{\text{cal}}{\text{deg.mole}}$$

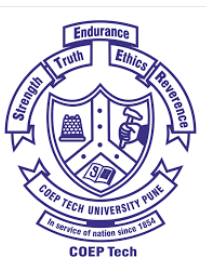
$$S_{298, \text{Cr}}^0 = 5.68 \frac{\text{cal}}{\text{deg.mole}}$$

$$S_{298, \text{CO}}^0 = 47.3 \frac{\text{cal}}{\text{deg.mole}}$$



Problem

Consider a 3-step cyclic process. The working substance is 1 kg-mol of a diatomic ideal gas initially at 300 K and 1 atm. The gas is heated at constant volume to 500 K, then expanded adiabatically to the initial temperature of 300 K, and finally compressed isothermally to the initial pressure of 1 atm. Each step occurs reversibly. Calculate ΔS for each step of the three-step cyclic process of expansion/compression of the ideal gas. ($\gamma = 7/5$, q for each step = 4157, 0, -3185 kJ/mol)



Problem

Calculate the standard entropy of solid copper at 1073°C (1346 K) from the following data:

$$S_{300 <\text{Cu}>} = 8 \text{ cal/deg/mole } (33.47 \text{ J/K/mol})$$

$$C_{p <\text{Cu}>} = 5.41 + 1.50 \times 10^{-3} T \text{ cal/deg/mole} \\ (22.63 + 6.27 \times 10^{-3} T \text{ J/K/mol}).$$



Problem

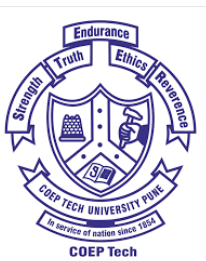
Zinc melts at 420°C (693 K) and its standard entropy at 25°C (298 K) is 9.95 cal/deg/mole (41.63 J/K/mol). Calculate the standard entropy of zinc at 750°C (1023 K).

Given: Heat of fusion of Zn at the melting point,

$$\Delta H_f = 1.74 \text{ kcal/mol e (7.28 kJ/mol)}.$$

$$C_{p, \text{Zn (s)}} = 5.35 + 2.40 \times 10^{-3} T \text{ cal/deg/mole} \\ (22.38 + 10.04 \times 10^{-3} T \text{ J/K/mol}).$$

$$C_{p, \text{Zn (l)}} = 7.50 \text{ cal/deg/mole (31.38 J/K/mol)}.$$



Problem

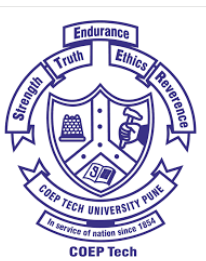
Calculate the entropy change of the Universe in the isothermal freezing of 1 g-mole of supercooled liquid gold at 1250 K, from the following data for gold.

$$T_m = 1336 \text{ K}$$

$$\Delta H_m = 12.36 \times 10^3 \text{ J/mol (at 1336 K)}$$

$$C_p(s) = 23.68 + 5.19 \times 10^{-3}T \text{ J/mol/K}$$

$$C_p(l) = 29.69 \text{ J/mol/K}$$



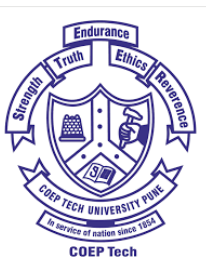
Free Energy

- The energy available for work
- The energy that can be easily released
- Unit of $T^*S = ???$
- Helmholtz Free Energy ($A = U - TS$)
- Gibbs Free Energy ($G = H - TS$)



Maxwell's Relationship

- $dU = TdS - PdV \Rightarrow \left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$
- $dH = TdS + VdP \Rightarrow \left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$
- $dA = -SdT - PdV \Rightarrow \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$
- $dG = -SdT + VdP \Rightarrow -\left(\frac{\partial S}{\partial P}\right)_T = \left(\frac{\partial V}{\partial T}\right)_P$



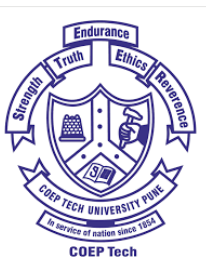
Gibbs Free Energy

- At constant temperature, $\Delta G = \Delta H - T\Delta S$
- ΔG should always be negative
- Driving force of any reaction is $\Delta G = \Delta H - T\Delta S$
- More negative of this factor => more driving force => more spontaneity



Gibbs – Helmholtz (G-H) Equation

$$\left(\frac{\partial(G/T)}{\partial T} \right)_P = - \frac{H}{T^2}$$
$$\left(\frac{\partial(\Delta G/T)}{\partial T} \right)_P = - \frac{\Delta H}{T^2}$$



Clapeyron Equation

$$\left(\frac{dT}{dP}\right)_{eq} = \frac{T \cdot \Delta V}{\Delta H_{trans}}$$



Clausius-Clapeyron Equation

- Liquid-vapour equilibria or Vaporisation

$$\frac{d(\ln P)}{dT} = \frac{\Delta H}{RT^2}$$



Free Energy as a Criterion for Equilibrium

- Spontaneity and Equilibrium
- $\Delta G = \Delta H - T\Delta S$
- Relation between free energy and entropy
- Closed system & Irreversible change => Spontaneous
- Closed system & Reversible change => Equilibrium

Significance

- $\Delta G = 0$ => equilibrium system
- $\Delta G < 0$ => spontaneous process
- $\Delta G > 0$ => reaction will spontaneously occur in the opposite direction



Statistical Nature of Entropy

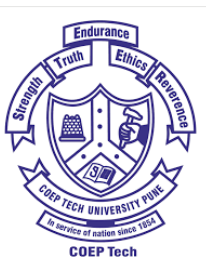
- Statistical entropy
- Ludwig Boltzmann by 1870
- In classical thermodynamics, we deal with *Macrostates* of substances
- In statistical thermodynamics, each microstate consists of a very large number of *Microstates* or quantum states (W)
- The largest possible number of arrangements of particles (W) amongst the different discrete energy levels of the system
- At equilibrium, $W = W_{max}$



Statistical Nature of Entropy (Contd...)

- The microscopic state of a system, which consists of a great number of atoms and molecules distributed in a space, is defined by the arrangements of the atoms/molecules within the space, the vibration of the atoms/molecules, the state of rotation, and the positions of electrons
- If the degree of freedom of the arrangements of the atoms or molecules, the state of vibration, the state of rotation, or the positions of the electrons is increased, W increases and so S also increases
- Hence, entropy, S , can be regarded as the degree of randomness of a system

From Prof. Butee Sir's class



Statistical Nature of Entropy (Contd...)

- There are Avogadro's number of molecules present in 1 mol of gas
- $N_A = 6.023 \times 10^{23}$ molecules
- So, the molecules can be arranged in such a way
- The number of ways N particles can be arranged amongst various energy levels are: $W = \frac{N!}{\prod n_i!}$
- $N = n_0 + n_1 + n_2 + n_3 + \dots + n_r = \sum_i n_i$
- Use Stirling's approximation
- $\ln N! \cong N \ln N - N$

$$\Delta S = k_B \ln W = -R[X_A \ln X_A + X_B \ln X_B]; X_A = \frac{n_A}{N}$$



3rd Law of Thermodynamics

- The entropy of a pure, perfect and crystalline solid at absolute zero is zero
- The entropy of a system at absolute zero in a state of thermodynamic equilibrium is zero
- To determine the absolute value of entropy (S) at any temperature
 - C_p s should be known throughout the range of temperatures
 - The substance is crystalline at absolute zero
- General Form
- Debye's equation



EXAMPLE 6.1 The equilibrium vapour pressure of ice at -10°C is 1.95 mmHg, and that of supercooled water at the same temperature is 2.149 mmHg. Calculate the free energy change accompanying the change of 1 mole of supercooled water to ice at -10°C .



EXAMPLE 6.2 Calculate the change of equilibrium transformation temperature of α -iron into γ -iron for raising pressure from 1 to 100 atm, with the help of the following data:

- (i) 910°C is the equilibrium transformation temperature for this at 1 atm pressure.
- (ii) At 910°C, $\Delta H_{\alpha \rightarrow \gamma}^0 = 912 \text{ J mol}^{-1}$, densities of α - and γ -iron are 7.571 and 7.633 g cm⁻³, respectively.
- (iii) Ignore change of volume with pressure.



EXAMPLE 6.4 The vapour pressure of liquid zinc above its melting point (692 K) is given by the approximate formula:

$$\ln p_{\text{Zn}}^0 = \frac{-14.765 \times 10^3}{T} + 12.805$$

The heat of fusion of zinc is equal to 6673 J mol^{-1} . Derive a formula for the vapour pressure over solid zinc below 692 K.

... with liquid zinc. Hence, at 692 K,



Example 4.1

The vapour pressure of liquid titanium at 2227°C (2500 K) is 1.503 mm Hg (200 N/m^2). The heat of vaporization at the normal boiling point of titanium is 104 kcal/mole (435.14 kJ/mol). Calculate its normal boiling point.

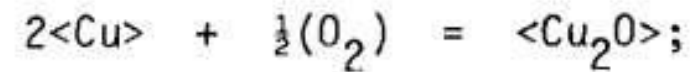


Example 4.C

The normal boiling point of zinc is 907°C (1180 K). Calculate its vapour pressure at 800°C (1073 K), assuming that zinc follows Trouton's rule.



Calculate the standard enthalpy and entropy changes at 25°C (298 K) for the reaction



$$\Delta G^0 = -40,500 - 3.92 T \log T + 29.5 T \text{ cal} \\ (-169,452 - 16.40 T \log T + 123.43 T J).$$

